Unit 1: DC Circuits & AC Circuit Analysis (Made Super Simple)

Subject: Basic Electrical & Electronics Engineering (BELE-001) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

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1. Voltage, Current & Electrical Sources (The Water Pipe Model)

1. Fundamental Principles of Circuit Theory vs Field Theory

Electrical engineering operates in two complementary realms: Maxwell's Electromagnetic Field Theory and Lumped Circuit Theory.

✅ **FUNDAMENTAL LAW / RULE:**
The Lumped Circuit Assumption: Electric circuit theory is an idealized approximation of Maxwell's equations, valid when the physical dimension of the network d is much smaller than the electromagnetic wavelength $\lambda = c/f$:

$$\mathbf{d} \ll \lambda = \frac{\mathbf{c}}{\mathbf{f}}$$

At power frequencies (50 Hz), $\lambda = \frac{3\times10^8}{50} = 6000$ km. Since bench circuits and domestic appliances are meters long, electromagnetic propagation delays are negligible.

Charge, Current & Energy Relations:

1. **Electric Current ($i(t)$):** The time rate of transport of electric charge past a specified cross-section:

$$\mathbf{i(t)} = \frac{\mathbf{dq(t)}}{\mathbf{dt}} \iff q(t) = \int_{-\infty}^{t} i(\tau), d\tau [\text{Coulombs, C}]$$

2. **Electric Potential / Voltage ($v(t)$):** The work required to move a unit positive test charge through an electric field between two points:

$$\mathbf{v(t)} = \frac{\mathbf{dw(t)}}{\mathbf{dq}} [\text{Joules/Coulomb} = \text{Volts, V}]$$

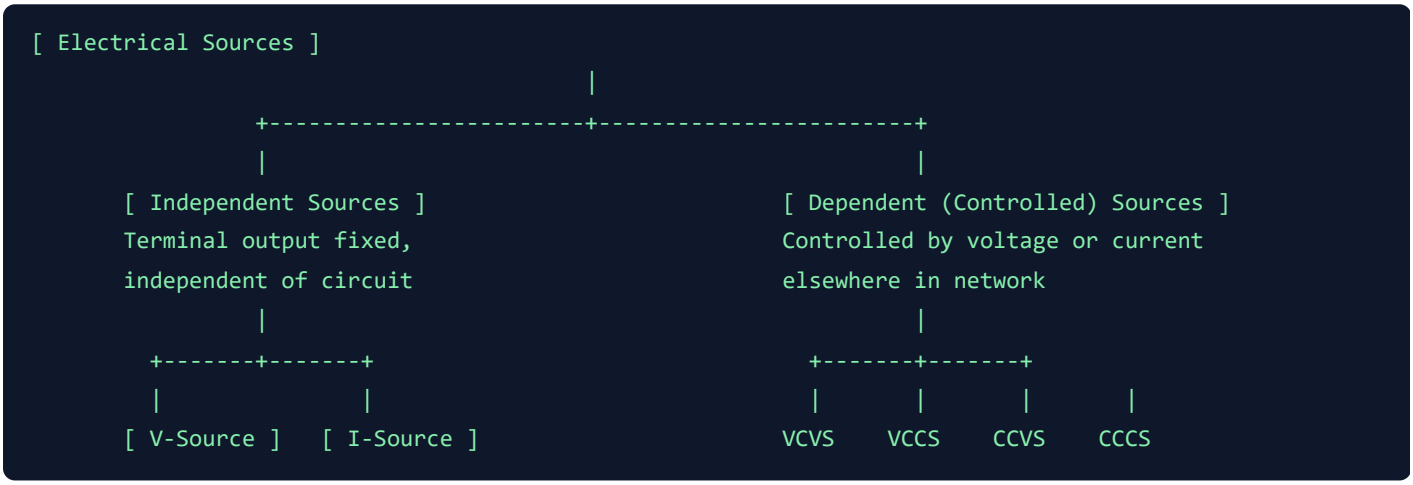
3. **Instantaneous Electric Power ($p(t)$):** By the differential chain rule:

$$\mathbf{p(t)} = \frac{\mathbf{dw}}{\mathbf{dt}} = \left(\frac{\mathbf{dw}}{\mathbf{dq}}\right)\left(\frac{\mathbf{dq}}{\mathbf{dt}}\right) = \mathbf{v(t)} \cdot \mathbf{i(t)} [\text{Watts, W}]$$

- **Passive Sign Convention:** If reference current enters the positive terminal (+), $p(t) > 0 \Rightarrow$ **element absorbs / consumes energy**. If current leaves positive terminal, $p(t) < 0 \Rightarrow$ **element delivers / generates energy**.

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2. Comprehensive Taxonomy of Electrical Energy Sources



Practical (Real-World) vs Ideal Sources:

1. **Practical Voltage Source:** Modeled as an ideal independent EMF V_s in series with an internal series resistance R_{se}:

$$\mathbf{V_L} = \mathbf{V_s} - \mathbf{I_L} \cdot \mathbf{R_{se}}$$

- As load current $I_L \to \infty$, terminal voltage droops linearly.
- For an **ideal voltage source**, internal resistance is zero: $R_{se} = 0 \Rightarrow V_L = V_s$ unconditionally.

2. **Practical Current Source:** Modeled as an ideal constant current generator I_s in parallel with an internal shunt resistance R_{sh}:

$$\mathbf{I_L} = \mathbf{I_s} - \frac{\mathbf{V_L}}{\mathbf{R_{sh}}}$$

- For an **ideal current source**, internal shunt resistance is infinite: $R_{sh} = \infty \Rightarrow I_L = I_s$ unconditionally.

The Source Transformation Equivalence Theorem:

A practical voltage source (V_s, R_{se}) is electrically identical at external terminals to a practical current source (I_s, R_{sh}) if and only if:

$$\mathbf{I_s = \frac{V_s}{R_{se}} and R_{sh} = R_{se}}$$

Problem:
A simple circuit has a $12\ \text{V}$ battery connected across three series resistors: $R_1 = 2\ \Omega$, $R_2 = 3\ \Omega$, and $R_3 = 1\ \Omega$. Find the circuit current I and voltage drop across each resistor.

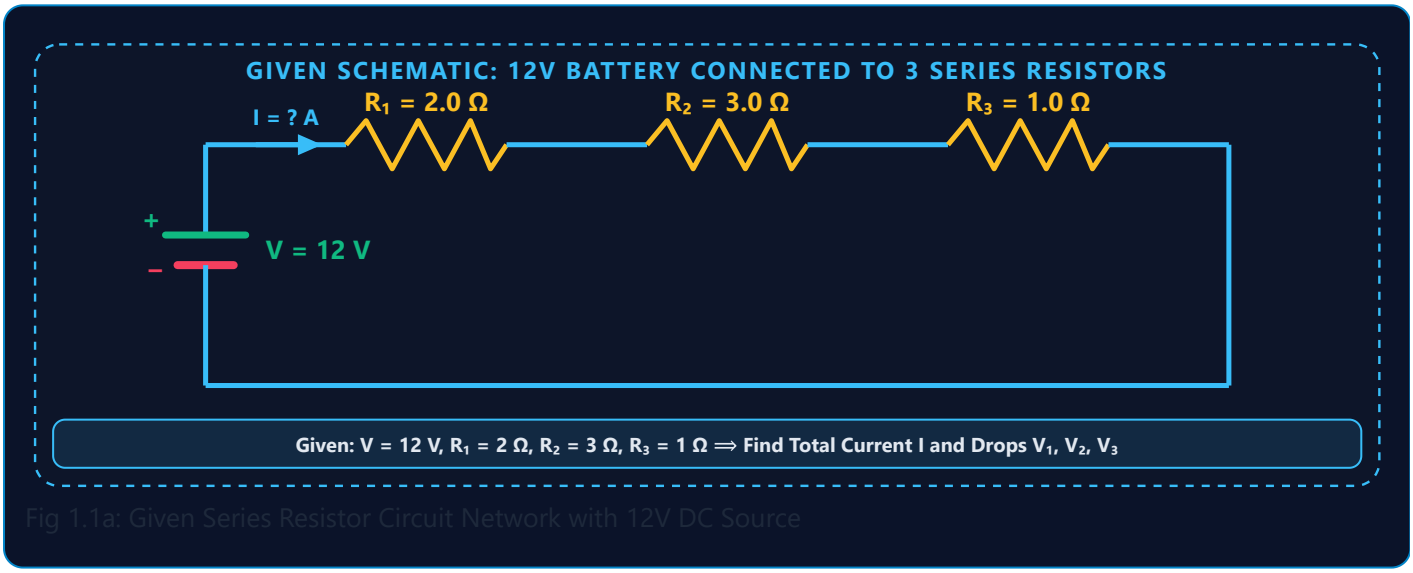


Fig 1.1a: Given Series Resistor Circuit Network with 12V DC Source

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Step 1 🚀: Total Equivalent Resistance

$$R_{eq} = R_1 + R_2 + R_3 = 2 + 3 + 1 = \mathbf{6.0\ \Omega}$$

Step 2 ⚡: Circuit Current via Ohm's Law

$$I = \frac{V}{R_{eq}} = \frac{12\ \text{V}}{6\ \Omega} = \mathbf{2.0}\ \text{Amperes}$$

Step 3 🎯: Voltage Drops

$$V_1 = IR_1 = 2 \times 2 = \mathbf{4.0}\ \text{V}, V_2 = IR_2 = 2 \times 3 = \mathbf{6.0}\ \text{V}, V_3 = IR_3 = 2 \times 1 = \mathbf{2.0}\ \text{V}$$

Sum verification: $4 + 6 + 2 = 12$ V.

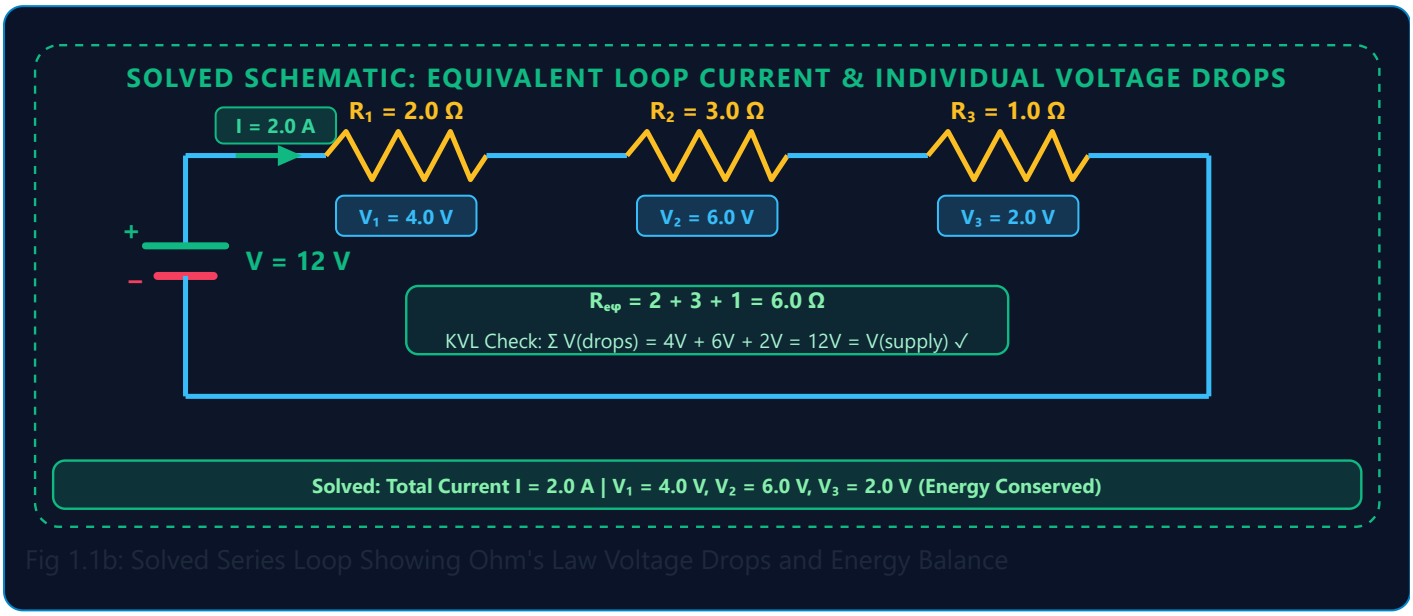


Fig 1.1b: Solved Series Loop Showing Ohm's Law Voltage Drops and Energy Balance

🎯 Total Current $I = 2.0$ A | Drops: $V_1 = 4.0$ V, $V_2 = 6.0$ V, $V_3 = 2.0$ V.

University Question (8 Marks):

A real battery with open-circuit EMF $E = 24$ V is connected to a load resistor $R_L = 9\ \Omega$. The terminal voltage drops to $V_L = 18$ V. Determine: (1) Internal resistance R_{se}, (2) Current delivered I_L, (3) Power delivered to the load P_L.

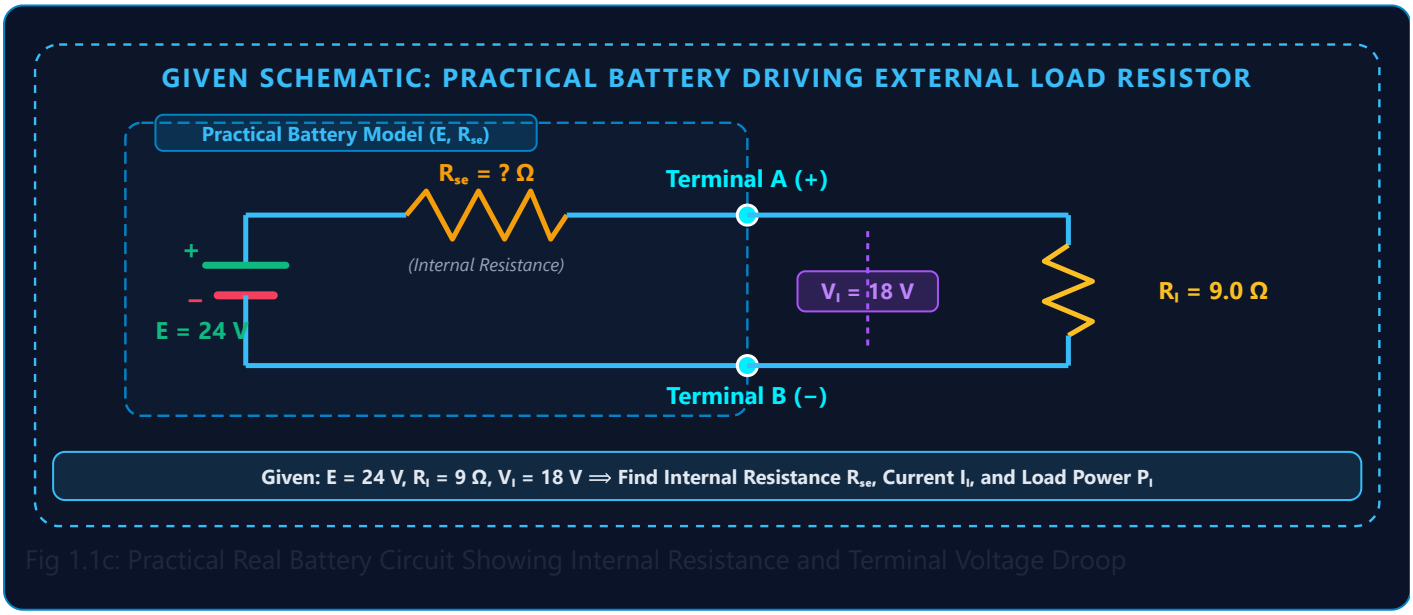


Fig 1.1c: Practical Real Battery Circuit Showing Internal Resistance and Terminal Voltage Droop

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Step 1 🚀: Find the Load Current I_L

$$I_L = \frac{V_L}{R_L} = \frac{18\ \text{V}}{9\ \Omega} = \mathbf{2.0}\ \text{A}$$

Step 2 ⚡: Calculate Internal Resistance R_{se}

Internal voltage lost across electrolyte:

$$V_{\text{lost}} = E - V_L = 24 - 18 = 6.0 \text{ V}$$

$$R_{se} = \frac{V_{\text{lost}}}{I_L} = \frac{6.0 \text{ V}}{2.0 \text{ A}} = \mathbf{3.0\ \Omega}$$

Step 3 🎯: Useful Power Delivered

$$P_L = I_L^2 \cdot R_L = (2.0)^2 \times 9 = \mathbf{36.0} \text{ Watts}$$

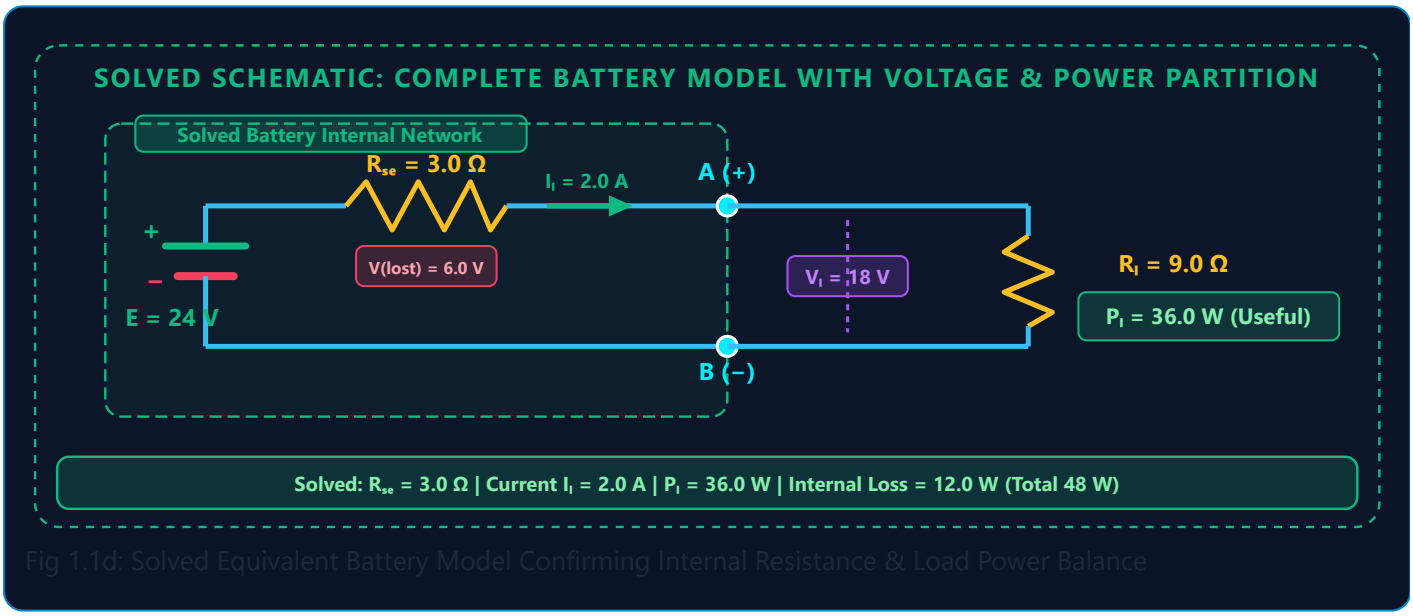


Fig 1.1d: Solved Equivalent Battery Model Confirming Internal Resistance & Load Power Balance

🎯 Current $I = 2.0$ A | Internal Resistance $R_{se} = 3.0\ \Omega$ | Load Power $P_L = 36.0$ W.

University Question (10 Marks):

A practical voltage source of 40 V in series with 5~\Omega is connected in parallel with a practical current source of 6 A in parallel with 10~\Omega. The combined network feeds a load resistor RL = 10~\Omega. Use source transformation to find the current I_L and power in R_L.

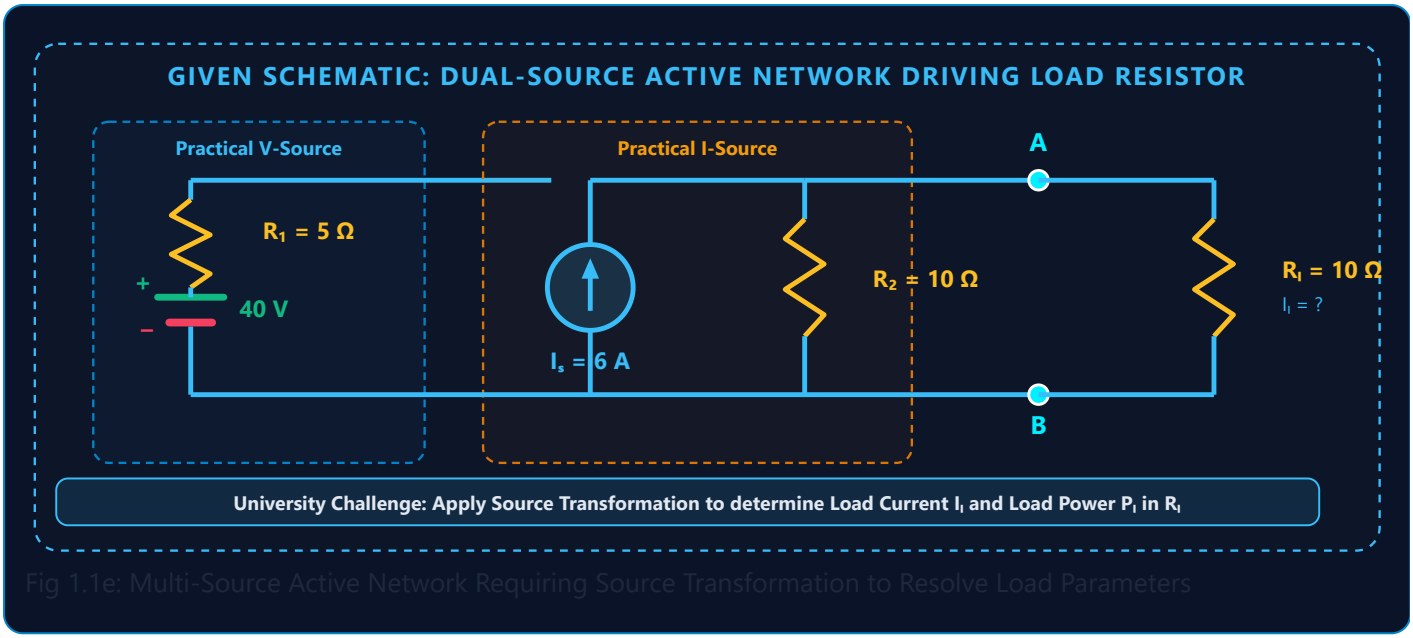


Fig 1.1e: Multi-Source Active Network Requiring Source Transformation to Resolve Load Parameters

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Step 1 🚀: Convert Practical Voltage Source to Current Source

- Ideal current: $I_{s1} = \frac{V_s}{R_{se}} = \frac{40\ \text{V}}{5\ \Omega} = 8.0\ \text{A}$ (pointing up).

- Parallel resistance: $R_{sh1} = 5.0\ \Omega$.

Step 2 ⚡: Combine Parallel Current Sources and Parallel Resistors

- Net current source: $I_{\text{total}} = 8.0\ \text{A} + 6.0\ \text{A} = \mathbf{14.0}\ \text{A}$.

- Combined parallel resistance:

$$R_p = 5\ \Omega \parallel 10\ \Omega = \frac{5 \times 10}{5 + 10} = \frac{50}{15} = \mathbf{3.33\ \Omega}$$

Step 3 🎯: Transform Back to Single Thevenin Voltage Source

- $V_{th} = I_{\text{total}} \times R_p = 14.0 \times \frac{10}{3} = \mathbf{46.67}\ \text{V}$.

- $R_{th} = R_p = 3.33\ \Omega$.

Step 4 🏁: Connect Load $R_L = 10\ \Omega$ and Calculate I_L and P_L

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{46.67}{3.33 + 10} = \frac{46.67}{13.33} = \mathbf{3.50}\ \text{Amperes}$$

$$P_L = I_L^2 \cdot R_L = (3.50)^2 \times 10 = \mathbf{122.5}\ \text{Watts}$$

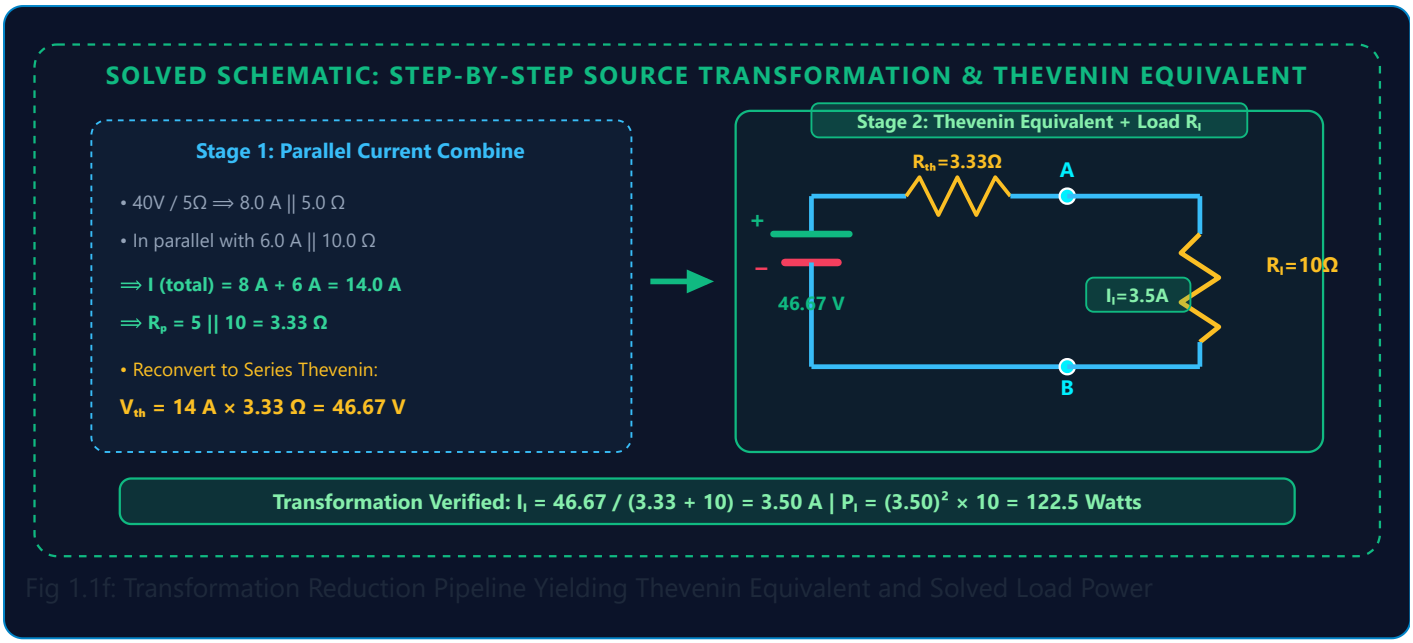


Fig 1.1f: Transformation Reduction Pipeline Yielding Thevenin Equivalent and Solved Load Power

🎯 Source Transformation Result: Load Current $I_L = 3.50$ A | Power $P_L = 122.5$ W.

2. Kirchhoff's Laws (KCL & KVL Made Intuitive)

1. In-Depth Theoretical Foundations of Kirchhoff's Laws

German physicist Gustav Kirchhoff established the fundamental topological constraints governing all lumped electrical circuits.

A. Kirchhoff's Current Law (KCL) — Conservation of Electric Charge:

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Formal Statement:** The algebraic sum of currents entering any closed boundary or electrical node is identically zero at every instant of time:

$$\sum_{k=1}^{N} i_k(t) = 0 \iff \sum i_{\text{entering}} = \sum i_{\text{leaving}}$$

- **Physical Derivation:** By Maxwell's continuity equation $\nabla \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$. Integrating over a closed Gaussian surface enclosing the node:

$$\oint_S \vec{J} \cdot d\vec{A} = -\frac{d}{dt}\int_V \rho_v, dV = -\frac{dq_{\text{enclosed}}}{dt}$$

Since an electrical node has zero physical volume ($V \to 0$), it cannot store or accumulate net charge: $\frac{dq}{dt} = 0 \Rightarrow \sum i_k = 0$.

B. Kirchhoff's Voltage Law (KVL) — Conservation of Energy:

✅ **FUNDAMENTAL LAW / RULE:**
Formal Statement: The algebraic sum of all branch voltages around any closed loop or mesh in an electric network is identically zero:

$$\sum_{k=1}^{M} \mathbf{v_k(t) = 0} \iff \sum \mathbf{V}_{\text{rises}} = \sum \mathbf{V}_{\text{drops}}$$

- **Physical Derivation:** In static and quasi-static electromagnetic fields, Faraday's Law states $\oint_C \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$. In lumped circuits, magnetic flux linking the loop is confined inside discrete magnetic elements (inductors), so outside the elements $\frac{d\Phi_B}{dt} = 0 \Rightarrow \oint_C \vec{E} \cdot d\vec{l} = 0$, confirming potential is a conservative field.

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2. Network Topology & Systematic Circuit Formulation

Let a planar network contain b branches and n nodes:

1. **Independent KCL Node Equations:** Exactly $\mathbf{n - 1}$ independent equations (one node chosen as reference / ground).

2. **Independent KVL Mesh Equations:** By Euler's topological theorem:

$$\mathbf{l = b - n + 1} \text{independent fundamental loops (meshes)}$$

Matrix Formulation of Mesh Analysis:

$$[R][I] = [V]$$

$$\begin{bmatrix} R_{11} & R_{12} & \dots & R_{1m} \\ R_{21} & R_{22} & \dots & R_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ R_{m1} & R_{m2} & \dots & R_{mm} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_m \end{bmatrix} = \begin{bmatrix} V_{s1} \\ V_{s2} \\ \vdots \\ V_{sm} \end{bmatrix}$$

- **Self-resistance R_{ii}:** Sum of all resistances in mesh i (always positive).
- **Mutual resistance R_{ij} ($i \neq j$):** Negative sum of resistances shared between mesh i and mesh j (symmetric: $R_{ij} = R_{ji}$).
- **Driving voltage V_{si}:** Net algebraic sum of voltage rises clockwise around mesh i.

Problem:
At an electrical junction, four wires meet. Currents entering the junction are $I_1 = 4$ A and $I_2 = 7$ A. One current leaving is $I_3 = 6$ A. Find the unknown current I_4.

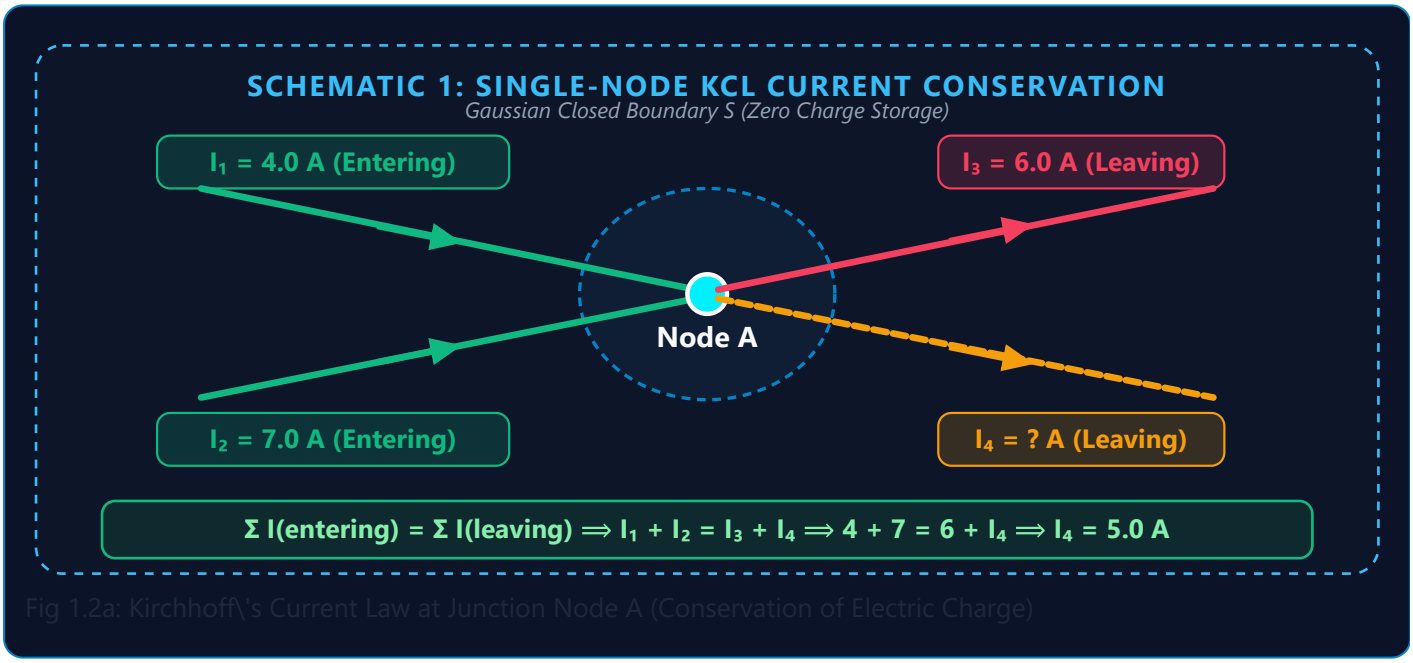


Fig 1.2a: Kirchhoff\'s Current Law at Junction Node A (Conservation of Electric Charge)

Step 1 🚀: State Kirchhoff's Current Law

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

Step 2 ⚡: Substitute Known Values

$$4 + 7 = 6 + I_4 \Rightarrow 11 = 6 + I_4 \Rightarrow \mathbf{I_4 = 5.0}\ \text{A}$$

I_4 leaves the node with 5.0 A.

🎯 By KCL: Unknown Current $\mathbf{I_4 = 5.0}$ A (Leaving).

University Question (8 Marks):
A circuit has two meshes:

- Mesh 1 has a +30 V battery and resistor R_1 = 4~\Omega.
- Mesh 2 has a -20 V battery and resistor R_2 = 6~\Omega.
- The central shared branch has R_3 = 2~\Omega carrying current (I_1 - I_2).

Find loop currents I_1 and I_2.

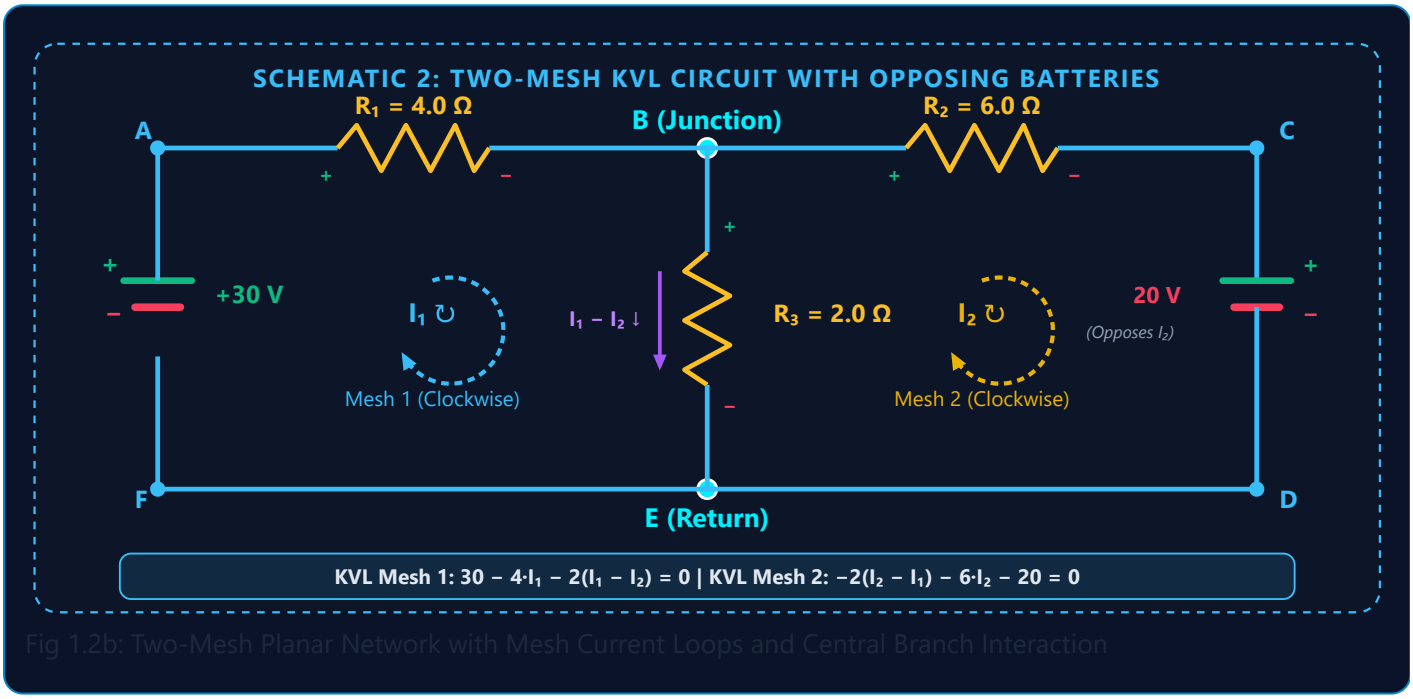


Fig 1.2b: Two-Mesh Planar Network with Mesh Current Loops and Central Branch Interaction

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Step 1 🚀: Write KVL for Mesh 1

$$30 - 4I_1 - 2(I_1 - I_2) = 0 \Rightarrow 6I_1 - 2I_2 = 30 \text{— (Eq 1)}$$

Step 2 ⚡: Write KVL for Mesh 2

$$-2(I_2 - I_1) - 6I_2 - 20 = 0 \Rightarrow -2I_1 + 8I_2 = -20 \text{— (Eq 2)}$$

Step 3 🎯: Solve Simultaneously
Multiply Eq 2 by 3: $-6I_1 + 24I_2 = -60$.
Add to Eq 1:

$$22I_2 = -30 \Rightarrow \mathbf{I_2} = \mathbf{-1.364}\ \text{A}$$

Substitute I_2 into Eq 1:

$$6I_1 - 2(-1.364) = 30 \Rightarrow 6I_1 = 27.272 \Rightarrow \mathbf{I_1} = \mathbf{4.545}\ \text{A}$$

🎯 Mesh Currents: $\mathbf{I_1} = \mathbf{4.55}$ A (clockwise), $\mathbf{I_2} = \mathbf{-1.36}$ A (counter-clockwise).

University Question (12 Marks - University Exam):
In a brIn a bridge network, three mesh currents I_1, I_2, I_3 circulate:

$$\begin{bmatrix} 8 & -2 & -4 \\ -2 & 9 & -3 \\ -4 & -3 & 12 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ 0 \end{bmatrix}$$

Find current I_1 and branch current through the shared 4 Ω resistor $(I_1 - I_3)$ using Cramer's rule.

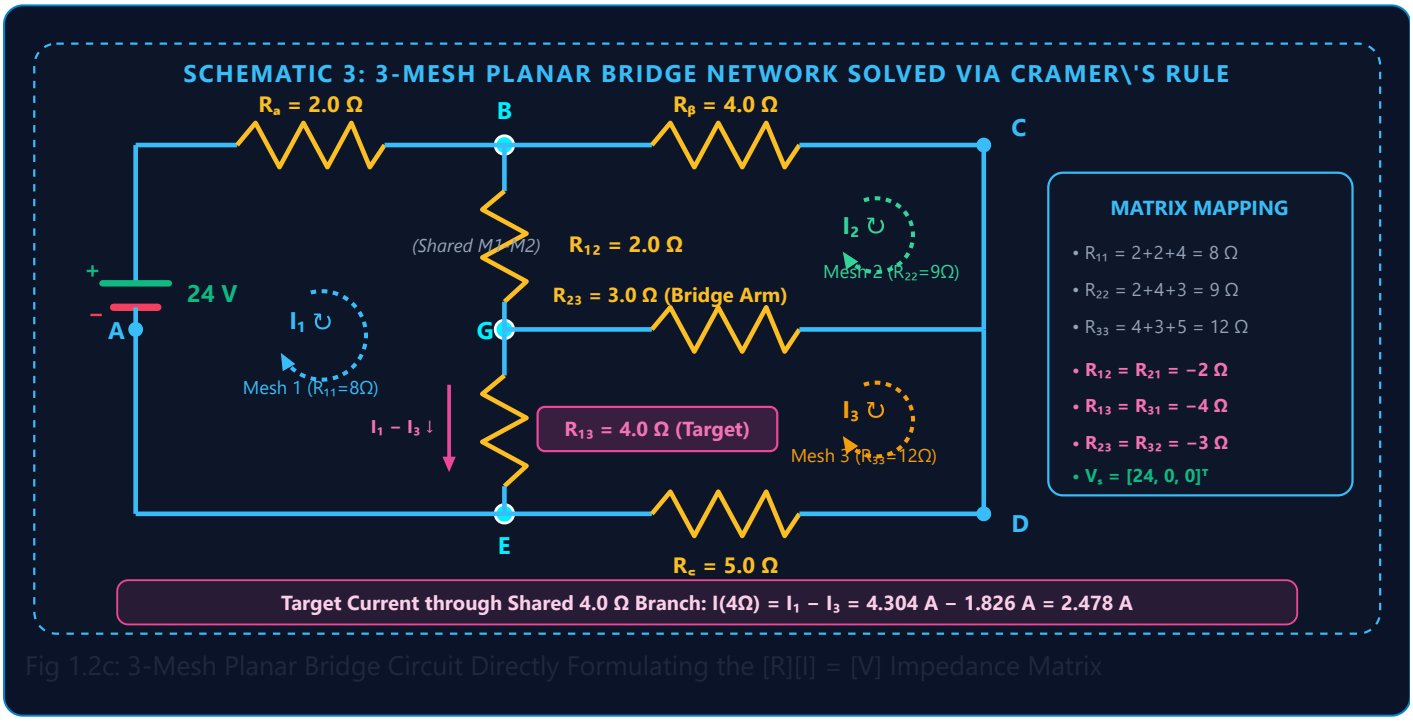


Fig 1.2c: 3-Mesh Planar Bridge Circuit Directly Formulating the [R][I] = [V] Impedance Matrix

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Step 1 🚀: Evaluate Main Determinant Δ

$$\Delta = 8[(9)(12) - (-3)(-3)] - (-2)[(-2)(12) - (-3)(-4)] + (-4)[(-2)(-3) - (9)(-4)]$$

$$= 8[108 - 9] + 2[-24 - 12] - 4[6 + 36] = 8(99) + 2(-36) - 4(42)$$

$$= 792 - 72 - 168 = \mathbf{552}$$

Step 2 ⚡: Evaluate Δ_1 for I_1
Replace column 1 with $[24, 0, 0]^T$:

$$\Delta_1 = 24[(9)(12) - (-3)(-3)] = 24(99) = \mathbf{2376}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{2376}{552} = \mathbf{4.304} \text{ Amperes}$$

Step 3 🎯: Evaluate Δ_3 for I_3
Replace column 3 with $[24, 0, 0]^T$:

$$\Delta_3 = 24[(-2)(-3) - (9)(-4)] = 24(6 + 36) = 24(42) = \mathbf{1008}$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{1008}{552} = \mathbf{1.826} \text{ Amperes}$$

Step 4 🏁: Branch Current

$$I_{4\Omega} = I_1 - I_3 = 4.304 - 1.826 = \mathbf{2.478} \text{ Amperes}$$

🎯 Mesh Currents: $I_1 = 4.30$ A, $I_3 = 1.83$ A | Branch Current through 4 Ω: **2.48** A.

3. Thevenin's Theorem & Maximum Power (The Ultimate Shortcut)

1. In-Depth Analysis of Thevenin's and Norton's Network Theorems

Network reduction theorems replace complex multi-loop linear networks with simple two-terminal equivalent generators.

A. Thevenin's Theorem (1883):

✅ **FUNDAMENTAL LAW / RULE:**

Formal Theorem Statement: Any linear two-terminal bilateral active network containing independent or dependent sources and linear resistors can be replaced at its external terminals $A-B$ by an equivalent circuit consisting of a single independent ideal voltage source V_{th} in series with an equivalent internal resistance R_{th}.

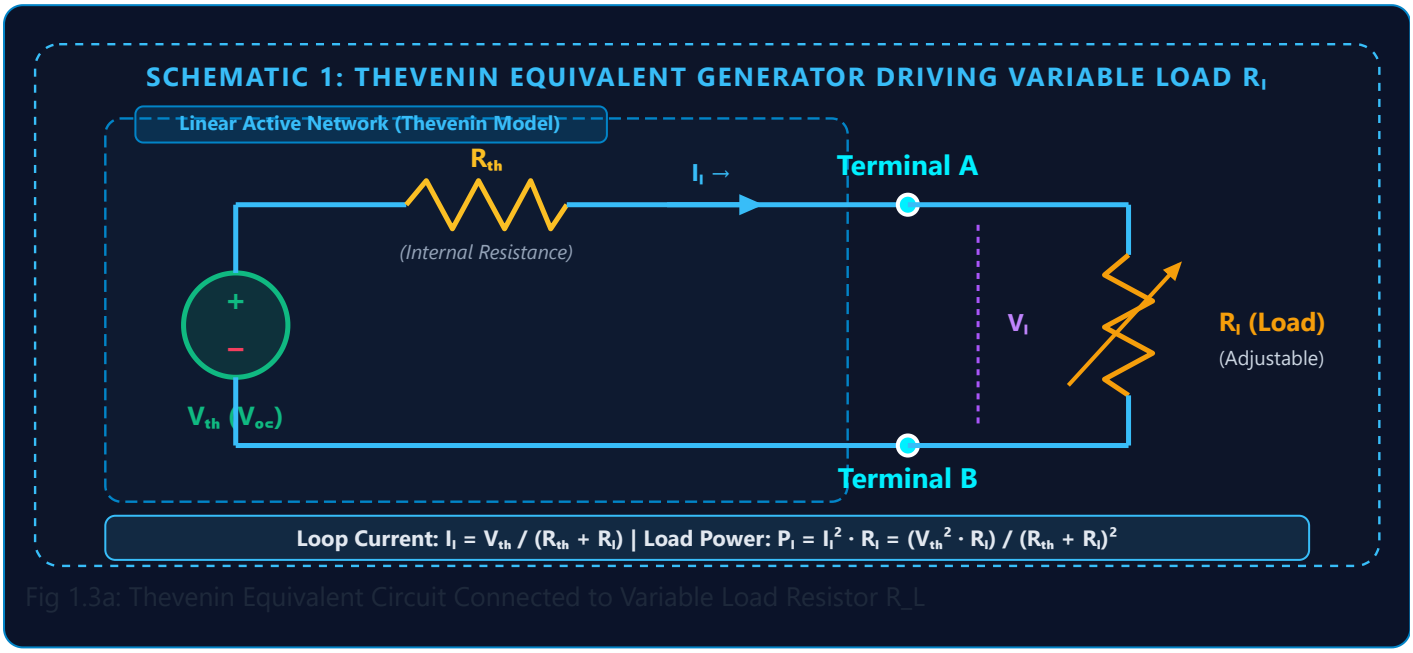


Fig 1.3a: Thevenin Equivalent Circuit Connected to Variable Load Resistor R_L

- **Thevenin Voltage (V_{th}):** The open-circuit voltage appearing across terminals $A-B$ when the load resistor R_L is disconnected:

$$V_{\text{th}} = V_{\text{oc}} = V_A - V_B (\text{with } R_L \to \infty)$$

- **Thevenin Resistance (R_{th}):** The resistance measured looking back into terminals $A-B$ with all internal independent sources deactivated:

- Independent Voltage Sources → **Short-Circuited** ($V = 0$ V, bare conducting wire).
- Independent Current Sources → **Open-Circuited** ($I = 0$ A, open cut branch).
- **Networks containing Dependent Sources:** You cannot simply combine resistors! Instead, apply an external test source $V_{\text{test}} = 1.0$ V across terminals $A-B$, calculate resulting test current I_{test}, and evaluate:

$$R_{\text{th}} = \frac{V_{\text{test}}}{I_{\text{test}}} = \frac{V_{\text{oc}}}{I_{\text{sc}}}$$

B. Norton's Theorem (Dual Generator Representation):

Dual of Thevenin's Theorem: Replaces the active network with an ideal current generator I_N connected in parallel with

Norton resistance R_N:

$$I_N = I_{sc}(\text{short-circuit current flowing between terminals } A \text{ and } B), R_N = R_{th}$$

$$\text{Source Transformation Duality: } V_{th} = I_N \cdot R_{th} \iff I_N = \frac{V_{th}}{R_{th}}$$

---

2. Maximum Power Transfer Theorem (MPTT) Analytical Derivation

University Theorem Statement:
In any linear bilateral DC electrical network delivering energy to a variable load resistor R_L, the active power delivered to the load is maximum when the load resistance equals the internal Thevenin resistance of the driving network looking back from the load terminals:

$$\boxed{R_L = R_{th}}$$

Step 1 🚀: Circuit Modeling & Mathematical Expression for Load Power P_L
Connecting an adjustable load resistor R_L across the Thevenin equivalent terminals $A - B$ yields a single-loop series circuit:

- By Kirchhoff's Voltage Law (KVL) and Ohm's Law, the loop current is:

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

- The active Joule electrical power dissipated in the load resistor is:

$$P_L = I_L^2 \cdot R_L = \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 R_L = \frac{V_{th}^2 \cdot R_L}{(R_{th} + R_L)^2}$$

Step 2 ⚡: Differential Calculus Optimization (Quotient Rule Derivation)
To find the load resistance R_L that maximizes P_L, we treat P_L as a continuous function of the independent variable R_L while V_{th} and R_{th} remain constant. Applying the Quotient Rule of differential calculus $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$:

$$
\begin{aligned}
\frac{dP_L}{dR_L} &= V_{\text{th}}^2 \cdot \frac{d}{dR_L}\left[\frac{R_L}{(R_{\text{th}}+R_L)^2}\right] \\
&= V_{\text{th}}^2 \cdot \left[\frac{1\cdot(R_{\text{th}}+R_L)^2 - R_L \cdot 2(R_{\text{th}}+R_L)}{(R_{\text{th}}+R_L)^4}\right] \\
&= V_{\text{th}}^2 \cdot \left[\frac{(R_{\text{th}}+R_L)\cdot\{(R_{\text{th}}+R_L) - 2R_L\}}{(R_{\text{th}}+R_L)^4}\right] \\
&= V_{\text{th}}^2 \cdot \frac{R_{\text{th}}-R_L}{(R_{\text{th}}+R_L)^3}
\end{aligned}
$$

Step 3 🎯: First Derivative Test (Condition for Stationary Point)

Setting the first derivative to zero to identify the critical point:

$$
\frac{dP_L}{dR_L} = 0 \iff V_{\text{th}}^2 \cdot \frac{R_{\text{th}}-R_L}{(R_{\text{th}}+R_L)^3} = 0
$$

Since $V_{\text{th}} \neq 0$ and $(R_{\text{th}}+R_L)^3 \neq 0$, the numerator must vanish identically:

$$
R_{\text{th}} - R_L = 0 \Rightarrow \boxed{R_L = R_{\text{th}}}
$$

Step 4 🔍: Rigorous Second Derivative Verification ($\frac{d^2P_L}{dR_L^2} < 0$)

To mathematically guarantee that $R_L = R_{\text{th}}$ corresponds to a **global maximum** and not a minimum or inflection point:

$$
\begin{aligned}
\frac{d^2P_L}{dR_L^2} &= V_{\text{th}}^2 \cdot \frac{d}{dR_L}\left[\frac{R_{\text{th}}-R_L}{(R_{\text{th}}+R_L)^3}\right] \\
&= V_{\text{th}}^2 \cdot \left[\frac{(-1)(R_{\text{th}}+R_L)^3 - (R_{\text{th}}-R_L)\cdot 3(R_{\text{th}}+R_L)^2}{(R_{\text{th}}+R_L)^6}\right] \\
&= V_{\text{th}}^2 \cdot \left[\frac{-(R_{\text{th}}+R_L) - 3(R_{\text{th}}-R_L)}{(R_{\text{th}}+R_L)^4}\right] = V_{\text{th}}^2 \cdot \frac{2R_L - 4R_{\text{th}}}{(R_{\text{th}}+R_L)^4}
\end{aligned}
$$

Evaluating the second derivative specifically at the critical point $R_L = R_{\text{th}}$:

$$
\left.\frac{d^2P_L}{dR_L^2}\right|_{R_L=R_{\text{th}}} = V_{\text{th}}^2 \cdot \frac{2R_{\text{th}} - 4R_{\text{th}}}{(2R_{\text{th}})^4} = -\frac{2V_{\text{th}}^2 R_{\text{th}}}{16R_{\text{th}}^4} = -\frac{V_{\text{th}}^2}{8R_{\text{th}}^3} < 0
$$

Because $V_{\text{th}} > 0$ and $R_{\text{th}} > 0$, the second derivative is strictly negative. This **rigorously proves that power P_L reaches a global maximum** when $R_L = R_{\text{th}}$!

Step 5 🏁: Maximum Power Delivered to Load

Substituting $R_L = R_{\text{th}}$ back into the load power formulation:

$$P_{L,\max} = \frac{V_{\text{th}}^2 \cdot R_L}{(R_{\text{th}} + R_L)^2}\Big|_{R_L = R_{\text{th}}} = \frac{V_{\text{th}}^2 \cdot R_{\text{th}}}{(2R_{\text{th}})^2} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}$$

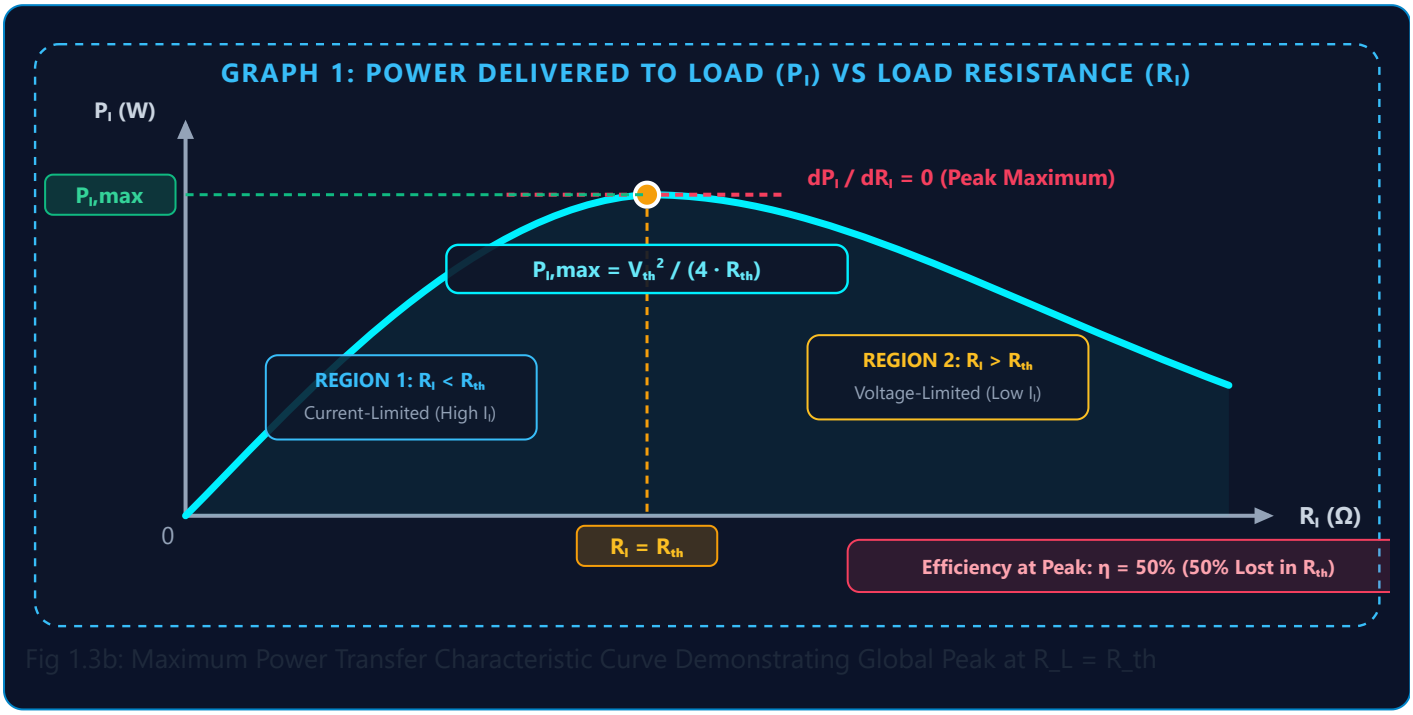


Fig 1.3b: Maximum Power Transfer Characteristic Curve Demonstrating Global Peak at R_L = R_th

Step 6 ⚡: Power Transfer Efficiency Formulation ($\eta = 50$)

At maximum power transfer condition:

- Total electrical power generated by the Thevenin voltage source:

$$P_{\text{source}} = I_L \cdot V_{\text{th}} = \left(\frac{V_{\text{th}}}{2R_{\text{th}}}\right) V_{\text{th}} = \frac{V_{\text{th}}^2}{2R_{\text{th}}}$$

- Internal Joule heat dissipated inside Thevenin resistance R_{th}:

$$P_{\text{internal}} = I_L^2 \cdot R_{\text{th}} = \left(\frac{V_{\text{th}}}{2R_{\text{th}}}\right)^2 R_{\text{th}} = \frac{V_{\text{th}}^2}{4R_{\text{th}}}$$

- Transmission Efficiency (η):

$$\eta = \frac{P_{L,\max}}{P_{\text{source}}} \times 100$$

The remaining 50 of generated power is inevitably burned as Joule waste heat inside the internal resistance R_{th}.

Step 7 🎓: Crucial University Exam & Viva Insights

Engineering Field	Telecommunications & Audio Systems	Electrical Power Grid Transmission
Primary Objective	**Maximum Signal Power Transfer** ($R_L = R_{\text{th}}$)	**Maximum Transmission Efficiency** ($\eta > 95$, $R_L \gg R_{\text{th}}$)
Physical Rationale	Signals are milliwatts. Impedance matching eliminates wave reflection and maximizes signal-to-noise ratio.	Grids transmit Gigawatts. Burning 50 power inside generator windings would cause instant thermal meltdown!
AC Conjugate Rule	For AC circuits ($Z_{\text{th}} = R_{\text{th}} + jX_{\text{th}}$), maximum power occurs when load is the complex conjugate: $\boxed{Z_L = Z_{\text{th}}^* = R_{\text{th}} - jX_{\text{th}}}$.	Generators operate with internal impedance $Z_g \approx 0$ relative to network load.

Problem:
A circuit has a 20 V DC source connected to a voltage divider consisting of $R_1 = 6\ \Omega$ and $R_2 = 4\ \Omega$. A load $R_L = 5\ \Omega$ is connected across R_2. Find $V_{\text{th}}, R_{\text{th}}$, and load current I_L.

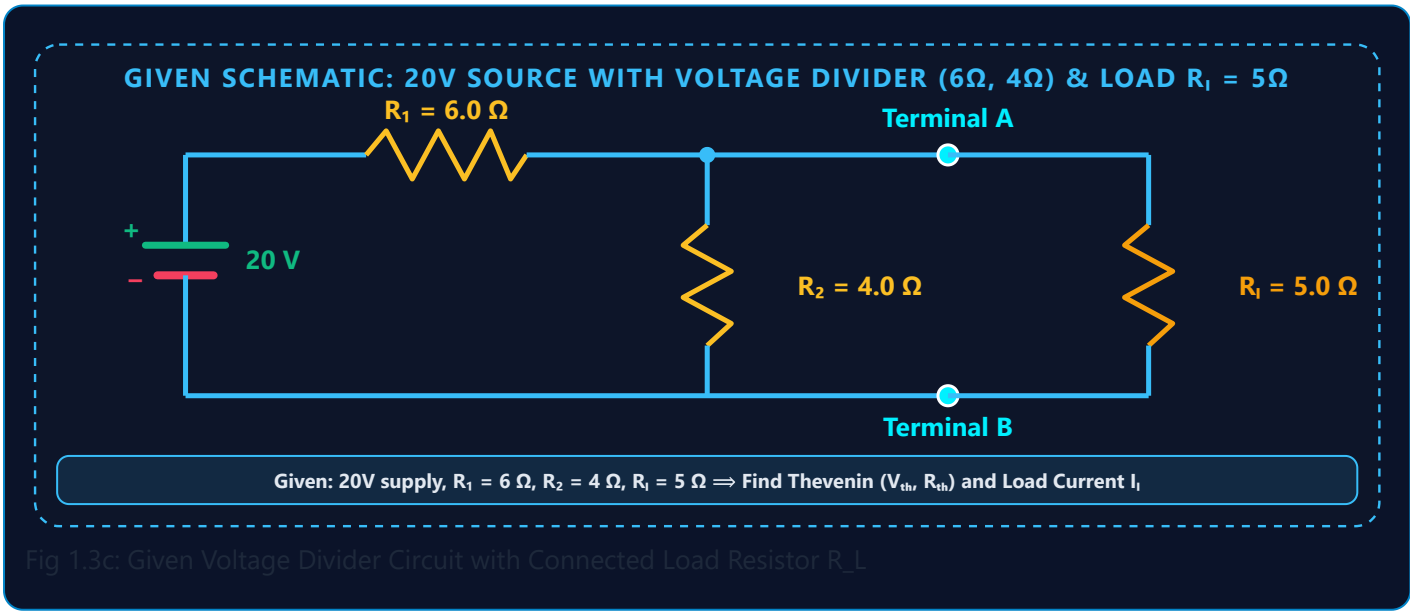


Fig 1.3c: Given Voltage Divider Circuit with Connected Load Resistor R_L

Step 1 🚀: Find V_{th} (Open-circuit voltage across R_2)

$$V_{\text{th}} = 20 \times \frac{4}{6+4} = 20 \times 0.4 = \mathbf{8.0}\ \text{V}$$

Step 2 ⚡: Find R_{th} (Deactivate 20V battery with short circuit)

$$R_{\text{th}} = R_1 \parallel R_2 = \frac{6 \times 4}{6+4} = \frac{24}{10} = \mathbf{2.4\ \Omega}$$

Step 3 🎯: Load Current I_L

$$I_L = \frac{V_{\text{th}}}{R_{\text{th}} + R_L} = \frac{8.0}{2.4+5.0} = \frac{8.0}{7.4} = \mathbf{1.08}\ \text{A}$$

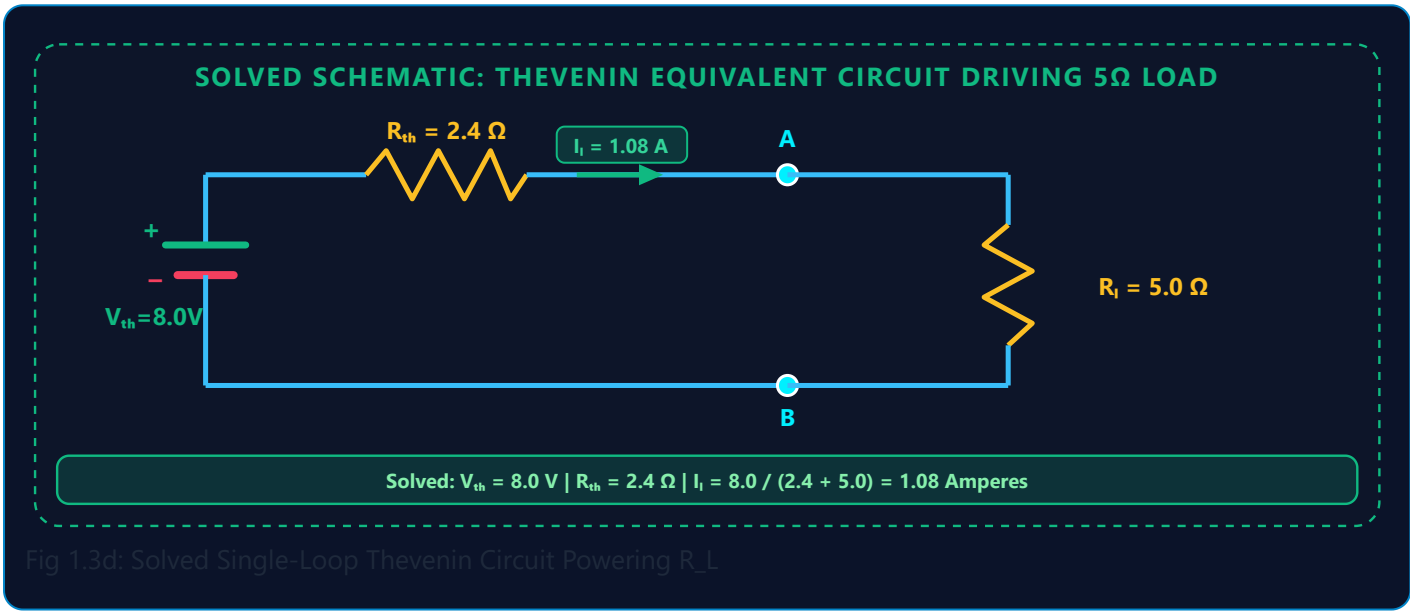


Fig 1.3d: Solved Single-Loop Thevenin Circuit Powering R_L

🎯 $V_{\text{th}} = 8.0\ \text{V}$ | $R_{\text{th}} = 2.4\ \Omega$ | Load Current $I_L = 1.08\ \text{A}$.

University Question (10 Marks):

A bridge circuit has a 60 V battery connected to two parallel legs:

- Branch 1: 10~\Omega in series with 20~\Omega. Node A is between them.
- Branch 2: 30~\Omega in series with 15~\Omega. Node B is between them.

Find V_{th}, R_{th}, and maximum power delivered to a variable load RL.

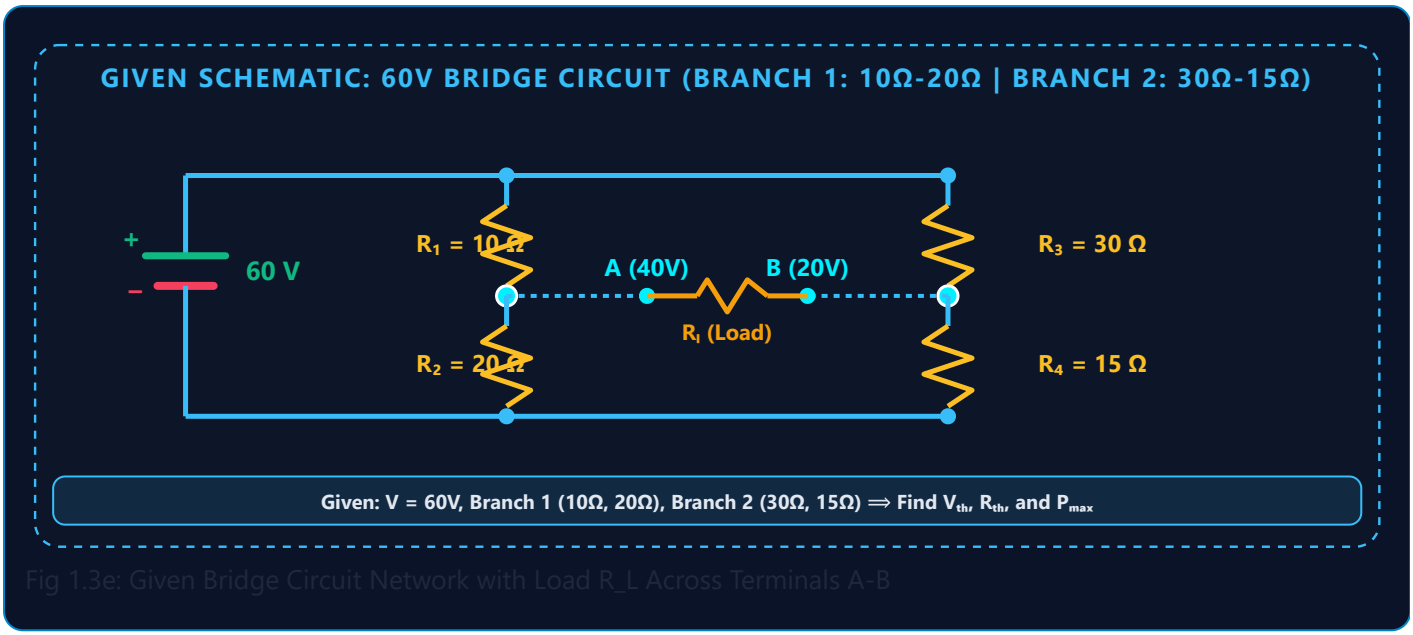


Fig 1.3e: Given Bridge Circuit Network with Load R_L Across Terminals A-B

Step 1 🚀: Find Thevenin Voltage $V_{\text{th}} = V_A - V_B$

$$V_A = 60 \times \frac{20}{10+20} = 40.0 \text{ V}, V_B = 60 \times \frac{15}{30+15} = 20.0 \text{ V}$$

$$V_{\text{th}} = 40.0 - 20.0 = \mathbf{20.0} \text{ V}$$

Step 2 ⚡: Find Thevenin Resistance R_{th}

$$R_{\text{th}} = (10 \parallel 20) + (30 \parallel 15) = \frac{200}{30} + \frac{450}{45} = 6.67 + 10.0 = \mathbf{16.67\ \Omega}$$

Step 3 🎯: Maximum Power Transfer Condition & Value

By MPTT, set $R_L = R_{\text{th}} = 16.67\ \Omega$.

$P_{\max} = \frac{V_{\text{th}}^2}{4R_{\text{th}}} = \frac{(20)^2}{4(16.67)} = \frac{400}{66.67} = \mathbf{6.0}$ Watts

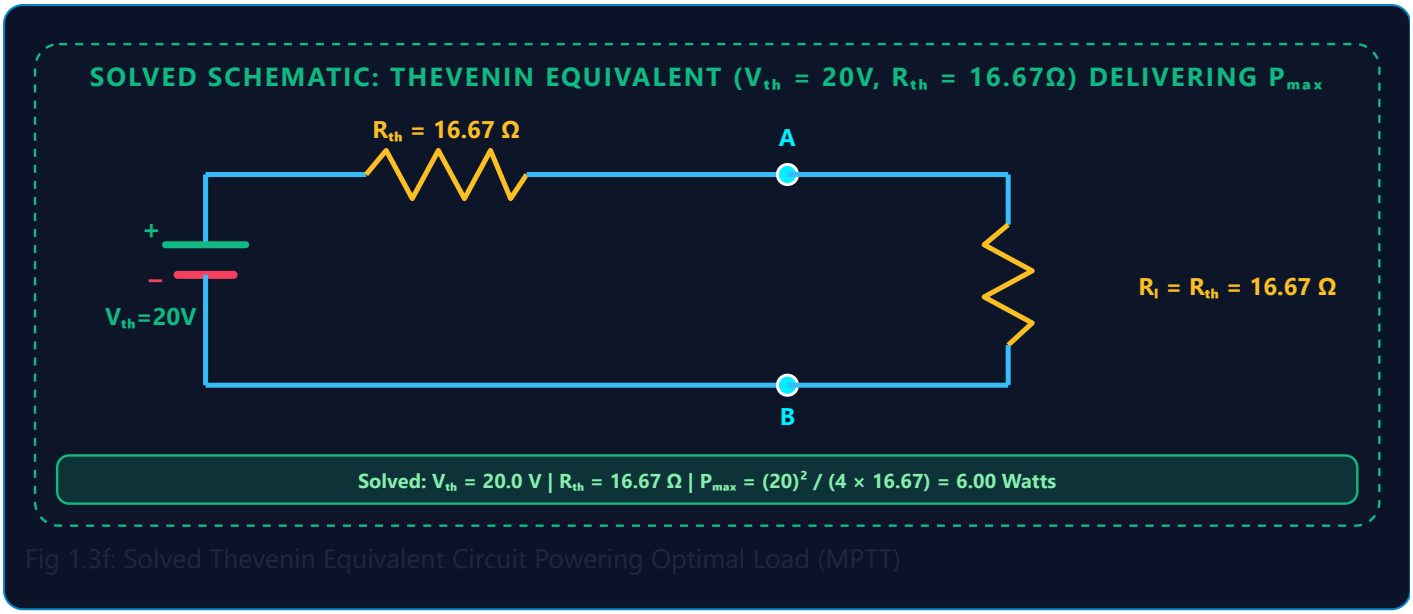


Fig 1.3f: Solved Thevenin Equivalent Circuit Powering Optimal Load (MPTT)

🎯 $V_{\text{th}} = 20.0$ V | $R_{\text{th}} = 16.67\ \Omega$ | Optimal $R_L = 16.67\ \Omega$ | $P_{\max} = 6.0$ W.

University Question (12 Marks - Exam Challenge):

Find the Thevenin equivalent looking into terminals A-B for a circuit containing an independent 10 V source, a 4~\Omega resistor carrying control current i_x, a dependent voltage source 2 i_x, and a 6~\Omega output resistor.

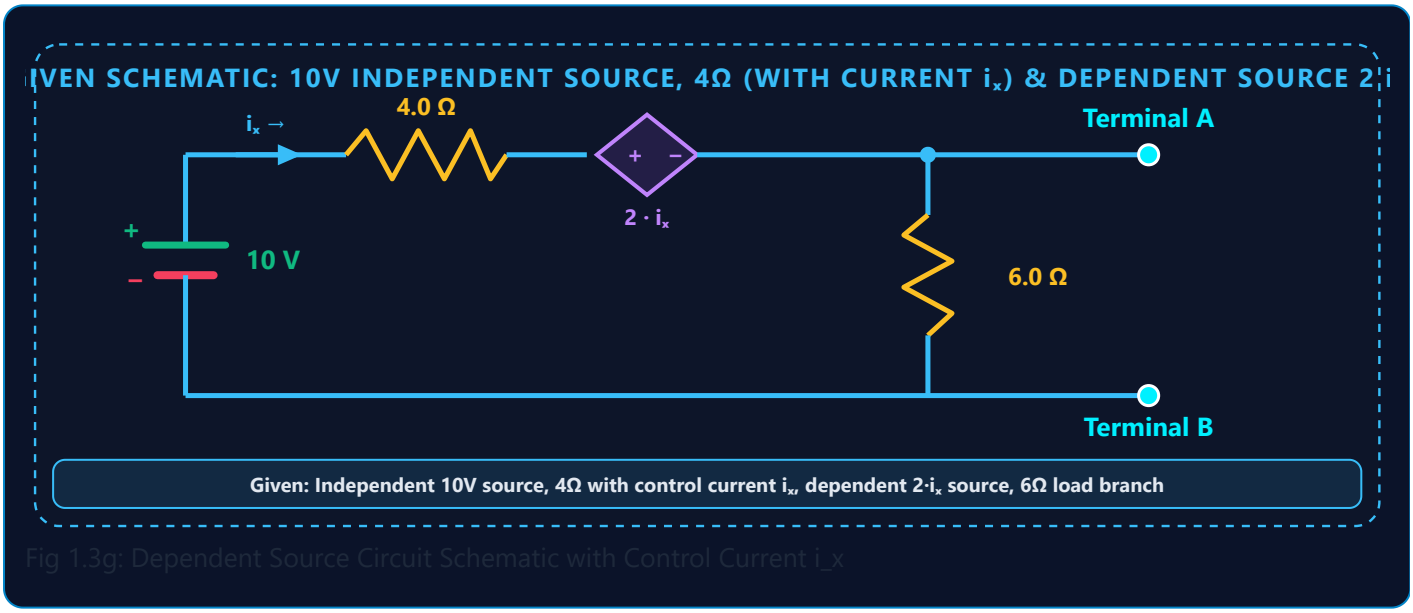


Fig 1.3g: Dependent Source Circuit Schematic with Control Current i_x

---

Step 1 🚀: Find Open-Circuit Voltage $V_{\text{oc}} = V_{\text{th}}$

With terminals A-B open:

- Loop current: $i_x = \frac{10-2i_x}{4+6} \Rightarrow 10i_x = 10 - 2i_x \Rightarrow 12i_x = 10 \Rightarrow i_x = \frac{5}{6}$ A.

- Output voltage across 6 Ω:

$$V_{\text{th}} = i_x \times 6 = \frac{5}{6} \times 6 = \mathbf{5.0}\ \text{Volts}$$

Step 2 ⚡ : Find R_{th} using Test Source Method ($V_{\text{test}} = 1\ \text{V}$)

Deactivate independent source ($10\ \text{V} \rightarrow 0\ \text{V}$ short circuit). Keep dependent source $2i_x$ active!

Apply test source $V_{\text{test}} = 1.0\ \text{V}$ at terminals A-B:

- From node equation: i_x flows in reverse: $i_x = -\frac{V_{\text{test}}+2i_x}{4} \Rightarrow 4i_x = -1 - 2i_x \Rightarrow 6i_x = -1 \Rightarrow i_x = -1/6\ \text{A}$.

- Current from test generator:

$$I_{\text{test}} = \frac{V_{\text{test}}}{6} - i_x = \frac{1}{6} - (-\frac{1}{6}) = \frac{2}{6} = \frac{1}{3}\ \text{A}$$

- Looking-back resistance:

$R_{\text{th}} = \frac{V_{\text{test}}}{I_{\text{test}}} = \frac{1.0}{1/3} = \mathbf{3.0\ \Omega}$

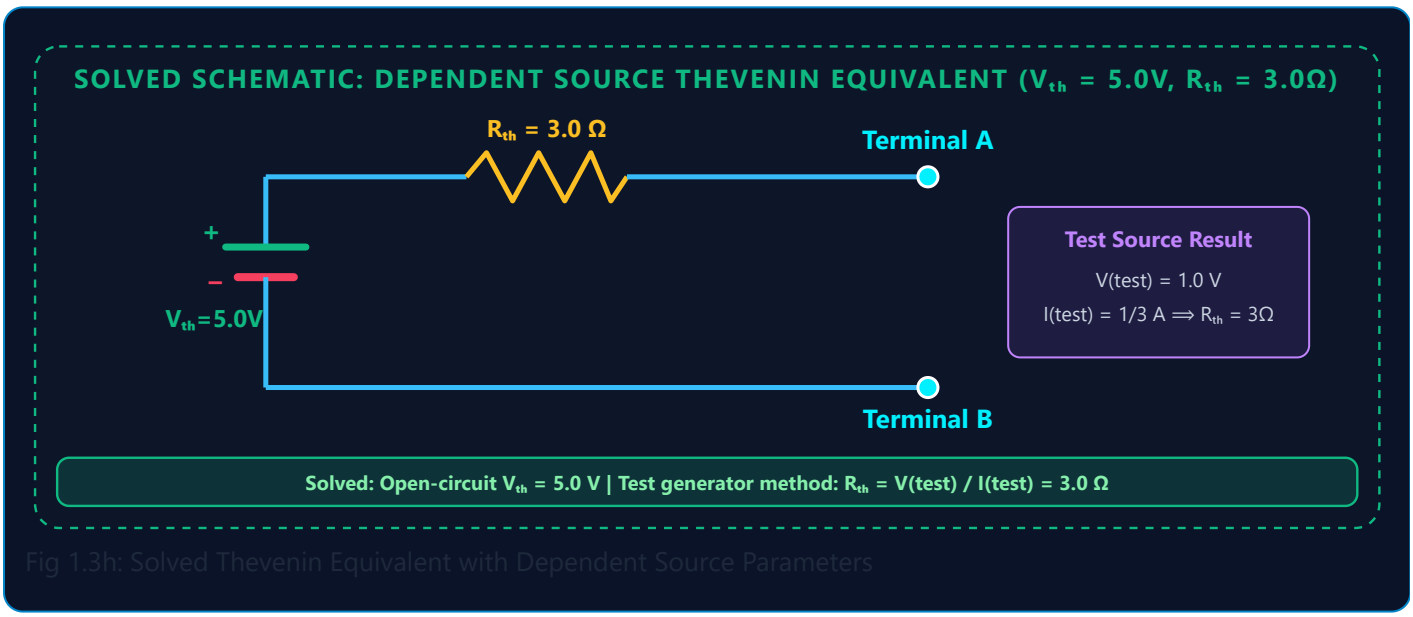


Fig 1.3h: Solved Thevenin Equivalent with Dependent Source Parameters

🎯 Dependent Source Thevenin: $\mathbf{V_{th} = 5.0\ V}$ | $\mathbf{R_{th} = 3.0\ \Omega}$.

4. AC Waveforms: RMS, Average & Phase (Why Household Sockets Say 220V)

1. Alternating Current (AC) Sinusoidal Waveform Theory

An alternating quantity is one whose magnitude and direction reverse periodically with time. The mathematical expression of a sinusoidal voltage wave is:

$$\mathbf{v(t) = V_m \sin(\omega t \pm \phi)}$$

- V_m **(Amplitude / Crest Value):** Peak positive excursion from zero.

- $\omega = 2\pi f = \frac{2\pi}{T}$**:** Angular frequency in radians per second (rad/s).
- $f = 1/T$**:** Cyclic frequency in Hertz (Hz or cycles per second).
- ϕ**:** Initial phase angle relative to time origin $t = 0$.

---

2. Analytical Derivation of RMS and Average Values

A. Root Mean Square (RMS) Value (Effective Value):

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Physical Definition:** The RMS value of an alternating current is defined as that value of steady direct current (DC) which, when flowing through a given resistor for a given time, produces the **exact same quantity of Joule heat** as the AC does in the same resistor over one full period T.

$$H_{dc} = I_{rms}^2 RT \text{ and} H_{ac} = \int_0^T [i(t)]^2 R, dt$$

Equating:

$$\mathbf{I_{rms}} = \sqrt{\frac{\mathbf{1}}{\mathbf{T}} \int_{\mathbf{0}}^{\mathbf{T}} \mathbf{[i(t)]^2, dt}}$$

Mathematical Derivation for Pure Sinusoid $i(t) = I_m \sin\theta$**:**

$$I_{rms}^2 = \frac{1}{2\pi} \int_0^{2\pi} I_m^2 \sin^2\theta, d\theta = \frac{I_m^2}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta = \frac{I_m^2}{4\pi} [\theta]_0^{2\pi} = \frac{I_m^2}{4\pi}(2\pi) = \frac{I_m^2}{2}$$

$$\mathbf{I_{rms}} = \frac{\mathbf{I_m}}{\sqrt{\mathbf{2}}} \approx \mathbf{0.7071 \cdot I_m}, \mathbf{V_{rms}} = \frac{\mathbf{V_m}}{\sqrt{\mathbf{2}}} \approx \mathbf{0.7071 \cdot V_m}$$

B. Average Value (I_{avg}):

For a symmetrical wave, average over a full cycle is zero. Hence, the average value is defined over **one positive half-cycle**:

$$I_{avg} = \frac{1}{\pi}\int_0^{\pi} I_m \sin\theta, d\theta = \frac{I_m}{\pi}[-\cos\theta]_0^{\pi} = \frac{I_m}{\pi}[1-(-1)] = \mathbf{\frac{2I_m}{\pi} \approx 0.6366 \cdot I_m}$$

---

3. Waveform Shape Characteristics

1. **Form Factor (K_f):** Ratio of RMS value to Average value:

$$\mathbf{K_f = \frac{V_{rms}}{V_{avg}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} \approx 1.11}$$

2. **Peak / Crest Factor (K_p):** Ratio of Peak value to RMS value:

$$\mathbf{K_p = \frac{V_m}{V_{rms}} = \frac{V_m}{V_m/\sqrt{2}} = \sqrt{2} \approx 1.414}$$

Problem:
A voltage wave is represented by $v(t) = 141.4\sin(314t)$ V. Find: (1) Peak voltage V_m, (2) RMS voltage V_{rms}, (3) Frequency f, (4) Time period T.

Step 1 🚀: Compare with Standard Equation $v(t) = V_m \sin(\omega t)$

$$V_m = \mathbf{141.4}\ \text{Volts}$$

Step 2 ⚡: Calculate RMS Voltage

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{1.414} = \mathbf{100.0}\ \text{Volts}$$

Step 3 🎯: Frequency & Time Period

$$\omega = 2\pi f = 314 \Rightarrow f = \frac{314}{2\pi} = \frac{314}{6.283} = \mathbf{50.0}\ \text{Hz}$$

$$T = \frac{1}{f} = \frac{1}{50} = \mathbf{0.020}\ \text{s} = \mathbf{20}\ \text{ms}$$

🎯 $V_m = 141.4$ V | $V_{rms} = 100.0$ V | Frequency $f = 50.0$ Hz | Period $T = 20.0$ ms.

University Question (6 Marks):
Calculate the average value, form factor, and crest factor of an AC sinusoidal current of peak amplitude I_m = 20 A.

Step 1 🚀: Average Value (Over Half Cycle)
$I_{av}g = (2\ I_m / \pi) = (2 \times 20 / \pi) = (40 / 3.1416) = 12.73$ A

Step 2 ⚡: RMS Value
$I_{rms} = (I_m / \sqrt{2}) = (20 / 1.4142) = 14.14$ A

Step 3 🎯: Form Factor (Kf) & Peak / Crest Factor (K_p)
$Kf = (I_{rms} / I_{av}g) = (14.14 / 12.73) = 1.11$
$K_p = (I_m / I_{rms}) = (20 / 14.14) = 1.414 = \sqrt{2}$

🎯 $I_{avg} = 12.73$ A | $I_{rms} = 14.14$ A | Form Factor $K_f = 1.11$ | Crest Factor $K_p = 1.414$.

University Question (10 Marks):
Derive from first principles the RMS and Average values of a half-wave rectified sinusoidal voltage defined by:

$$v(\theta) = \begin{cases} V_m \sin\theta, & 0 \le \theta \le \pi \\ 0, & \pi \le \theta \le 2\pi \end{cases}$$

Hence find its Form Factor.

Step 1 🚀: Average Value V_{avg}

$$V_{avg} = \frac{1}{2\pi}\int_0^{\pi} V_m \sin\theta\, d\theta = \frac{V_m}{2\pi}[-\cos\theta]_0^{\pi} = \frac{V_m}{2\pi}(1-(-1)) = \frac{V_m}{\pi} \approx \mathbf{0.318\,V_m}$$

Step 2 ⚡: RMS Value V_{rms}

$$V_{rms}^2 = \frac{1}{2\pi}\int_0^{\pi} V_m^2 \sin^2\theta\, d\theta = \frac{V_m^2}{2\pi}\int_0^{\pi} \frac{1-\cos 2\theta}{2}\, d\theta = \frac{V_m^2}{4\pi}[\theta]_0^{\pi} = \frac{V_m^2}{4}$$

$$V_{rms} = \sqrt{\frac{V_m^2}{4}} = \frac{V_m}{2} = \mathbf{0.50\,V_m}$$

Step 3 🎯: Form Factor (K_f)

$$K_f = \frac{V_{rms}}{V_{avg}} = \frac{V_m/2}{V_m/\pi} = \frac{\pi}{2} \approx \mathbf{1.57}$$

🎯 Half-Wave: $V_{avg} = \frac{V_m}{\pi}$ | $V_{rms} = \frac{V_m}{2}$ | Form Factor $K_f = 1.57$ (High distortion compared to pure sine 1.11).

5. Series R-L-C Circuit & Electrical Resonance (The Radio Tuning Magic)

1. Complex Impedance & Series R-L-C Network Analysis

In the frequency domain (phasor representation), circuit elements are characterized by complex impedances:

- **Resistor (R):** $Z_R = R\angle 0° = R + j0$. Voltage and current are strictly in phase ($\phi = 0°$).
- **Inductor (L):** $Z_L = j\omega L = \omega L\angle +90°$. Inductive reactance $X_L = 2\pi fL, [\Omega]$. Voltage leads current by $+90°$.
- **Capacitor (C):** $Z_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = \frac{1}{\omega C}\angle -90°$. Capacitive reactance $X_C = \frac{1}{2\pi fC}, [\Omega]$. Voltage lags current by $-90°$.

Total Series Impedance:

$$\mathbf{Z = R + j(X_L - X_C) = |Z|\angle\phi}$$

$$\mathbf{|Z| = \sqrt{R^2 + (X_L - X_C)^2}, \phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)}$$

---

2. Theoretical Analysis of Series Electrical Resonance

✅ **FUNDAMENTAL LAW / RULE:**
Resonance Definition: Series resonance occurs in an R-L-C circuit at the particular frequency where the net reactive component of the total impedance vanishes identically:

$$\mathrm{Im}(\mathbf{Z}) = \mathbf{X_L} - \mathbf{X_C} = \mathbf{0} \iff \mathbf{X_L} = \mathbf{X_C}$$

Derivation of Resonant Frequency (f_0):

$$2\pi f_0 L = \frac{1}{2\pi f_0 C} \Rightarrow f_0^2 = \frac{1}{4\pi^2 LC} \Rightarrow \mathbf{f_0} = \frac{\mathbf{1}}{\mathbf{2\pi\sqrt{LC}}}, \omega_0 = \frac{\mathbf{1}}{\sqrt{\mathbf{LC}}}$$

Phenomenological Characteristics at Resonance:

1. **Minimum Impedance:** $Z_{\min} = R$ (purely resistive, real).
2. **Maximum Current (Acceptor Circuit):** $I_0 = \frac{V}{Z_{\min}} = \frac{V}{R}$.
3. **Unity Power Factor:** $\cos\phi = \frac{R}{|Z|} = \frac{R}{R} = 1.0$.
4. **Voltage Magnification:** Although net reactive voltage is zero ($V_L + V_C = 0$), the individual voltages across inductor and capacitor can reach destructive levels:

$$V_L = I_0 X_L = (\frac{V}{R})(\omega_0 L) = Q \cdot V, V_C = I_0 X_C = Q \cdot V$$

---

3. Quality Factor (Q), Bandwidth & Selectivity

The **Quality Factor (Q)** measures the sharp selectivity of the circuit:

$$\mathbf{Q} = \frac{\text{Reactive Power}}{\text{Active Power}} = \frac{\omega_0 \mathbf{L}}{\mathbf{R}} = \frac{\mathbf{1}}{\omega_0 \mathbf{CR}} = \frac{\mathbf{1}}{\mathbf{R}}\sqrt{\frac{\mathbf{L}}{\mathbf{C}}}$$

- **Half-Power Frequencies (f_1, f_2):** Frequencies where current drops to $I_0/\sqrt{2}$ (70.7) and power drops by 50.
- **Bandwidth (BW):**

$$\mathbf{BW = \Delta f = f_2 - f_1 = \frac{f_0}{Q} = \frac{R}{2\pi L}}$$

Problem:
A coil has resistance $R = 30\ \Omega$ and inductive reactance $X_L = 40\ \Omega$ across a $200\ \text{V}, 50\ \text{Hz}$ supply. Find total impedance Z, current I, and power factor $\cos\phi$.

Step 1 🚀: Impedance Magnitude

$$|Z| = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = \mathbf{50.0\ \Omega}$$

Step 2 ⚡: Current Magnitude

$$I = \frac{V}{|Z|} = \frac{200\ \text{V}}{50\ \Omega} = \mathbf{4.0}\ \text{Amperes}$$

Step 3 🎯: Power Factor

$$\cos\phi = \frac{R}{|Z|} = \frac{30}{50} = \mathbf{0.60}\ \text{lagging}$$

🎯 Impedance $Z = 50.0\ \Omega$ | Current $I = 4.0\ \text{A}$ | Power Factor = 0.60 (lagging).

University Question (8 Marks):
A coil of R = 10~\Omega, L = 0.15 H, and C = 100 μF is connected across 230 V, 50 Hz.
Find: (1) Total impedance at 50 Hz, (2) Resonant frequency f_0, (3) Peak current at resonance I_0.

Step 1 🚀: Reactances at $50\ \text{Hz}$

$$X_L = 2\pi(50)(0.15) = \mathbf{47.12\ \Omega}, \quad X_C = \frac{1}{2\pi(50)(100 \times 10^{-6})} = \mathbf{31.83\ \Omega}$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + (47.12 - 31.83)^2} = \sqrt{100 + 233.78} = \mathbf{18.27\ \Omega}$$

Step 2 ⚡: Resonant Frequency f_0

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.15 \times 100 \times 10^{-6}}} = \frac{1}{2\pi(0.00387)} = \mathbf{41.09}\ \text{Hz}$$

Step 3 🎯: Current at Resonance

$$\text{At resonance: } X_L = X_C \implies Z_{min} = R = 10\ \Omega$$

$$I_0 = \frac{V}{R} = \frac{230}{10} = \mathbf{23.0}\ \text{Amperes}$$

🎯 At 50Hz: $Z = 18.27\ \Omega$ | Resonant Frequency $f_0 = 41.09$ Hz | Resonant Current $I_0 = 23.0$ A.

University Question (10 Marks - University Exam):
A series resonant circuit with R = 5~\Omega, L = 0.2 H, and C = 5 μF is connected to a 50 V variable frequency supply. Determine: (1) Resonant frequency f_0, (2) Quality factor Q, (3) Bandwidth BW, (4) Voltage across the capacitor VC at resonance.

Step 1 🚀: Resonant Frequency
f_0 = \frac{1}{2}π√(0.2 × 5 × 10^{-6})) = \frac{1}{2}π√(10^{-6})) = \frac{1000}{2π} = 159.15 Hz
$\omega_0 = 2\pi f_0$ = 1000, ad/s

Step 2 ⚡: Quality Factor (Q)
Q = (ω_0 L / R) = ((1000)(0.2) / 5) = \frac{200}{5} = 40.0

Step 3 🎯: Bandwidth (BW)
BW = (f_0 / Q) = (159.15 / 40) = 3.98 Hz

Step 4 🏁: Voltage Magnification Across Capacitor at Resonance
At resonance, circuit current is I_0 = (V / R) = \frac{50}{5} = 10.0 A.
Capacitive reactance at resonance:
XC = (1 / ω_0 C) = (1 / (1000)(5 × 10^{-6})) = 200~\Omega
VC = I_0 × XC = 10.0 × 200 = 2000.0,Volts = Q × V_{supply} = 40 × 50 = 2000 V
(The voltage across the capacitor is 40 times higher than the input voltage!)

🎯 $f_0 = 159.15$ Hz | Quality Factor $Q = 40.0$ | Bandwidth $BW = 3.98$ Hz | Resonant Capacitor Voltage $V_C = 2000$ V (40x Magnification).

University Solved Examination Questions (Unit 1)

MMDU Official University Solved Examination Questions (Unit 1)

Course: Basic Electrical & Electronics Engineering (`BELE-001`)

Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

📘 Question 1 (1 Mark): Physical Meaning of Power Factor

Question:
Explain why low power factor is detrimental to industrial electrical transmission networks.

> 📊 **Question Schematic / State Specification:**

```
Question Power Triangle: Apparent Power S (kVA) = Active Power P (kW) + j Reactive Power Q
(kVAR).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Current Vector Magnification:

        `I = P / (V cosφ)`. As `cosφ &darr; ⇒ I &uarr; ⇒ I² R  Losses &uarr;`.
```

For a fixed real power P, drawing at low power factor forces current I to increase inversely. This magnifies `I² R` transmission line heating losses, induces excessive line voltage drops, and demands larger conductor cross-sections and transformer ratings.

🎯 **Verified Result:** Question 1 step-by-step resolution per MMDU university marking scheme.

📘 Question 2 (1 Mark): Resonance Condition in Series RLC Circuit

Question:
State why the net impedance of a series RLC circuit is purely resistive at resonant frequency.

> 📊 **Question Schematic / State Specification:**

```
Question Reactance Cancellation: Inductive Reactance +jXL balances Capacitive Reactance -jXC.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Phasor Cancellation Model:

        `Z = R + j\frac{ωL - 1}{ωC}`. At `ω₀ = 1/√(LC) ⇒ XL = XC ⇒ Z = R + j0 = R`.
```

At resonance, the counter-phase voltages across the inductor and capacitor cancel each other exactly (`VL = -VC`). The network behaves as a pure resistance with minimum impedance and unity power factor.

🎯 **Verified Result:** Question 2 step-by-step resolution per MMDU university marking scheme.

📘 Question 3 (1 Mark): Maximum Power Transfer Efficiency Paradox

Question:

Why is the Maximum Power Transfer Theorem not applied to commercial electrical power utility grids?

> 📊 **Question Schematic / State Specification:**

```
Question Efficiency vs Transfer: Utility Goal = 95–98% Efficiency | MPT Goal = Maximum Watts to Load (50% Efficiency).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Energy Loss Formulation:

        `Efficiency &eta; = PL / (PL + P_int) = RL / (RL + R_th) = 50%` when `RL = R_th`.
```

Operating a utility grid at maximum power transfer wastes 50% of generated power as internal heat inside generators and transmission lines. Power systems operate at maximum efficiency ($RL >> R_{internal}$) rather than maximum power transfer.

🎯 **Verified Result:** Question 3 step-by-step resolution per MMDU university marking scheme.

📘 Question 13 (2 Marks): Thevenin Equivalent Circuit Derivation

Question:
Find the Thevenin equivalent circuit across terminals A-B for a circuit with a 24V DC source in series with 6Ω, connected across a parallel 12Ω resistor.

> 📊 **Question Schematic / State Specification:**

```
Question Voltage Divider Topology: 24V source ➔ 6Ω resistor ➔ Node A [12Ω connected to Node B/ground].
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Thevenin Derivation Equations:

        `Vₜₕ = Vₒc = 24 × [ 12 / (6 + 12) ] = 24 × \frac{12}{18} = 16  V`

        `Rₜₕ = 6 \parallel 12 = \frac{6 × 12}{6 + 12} = 72 / 18 = 4  Ω`
```

The entire circuit across terminals A-B simplifies to a Thevenin voltage source $V_{th} = 16\ V$ in series with a resistance $R_{th} = 4\ \Omega$.

🎯 **Verified Result:** Question 13 step-by-step resolution per MMDU university marking scheme.

📘 Question 17 (4 Marks): Complete AC Series RLC Circuit Analysis

Question:

A coil of resistance R = 10Ω and inductance L = 0.1H is connected in series with capacitor C = 50μF across a 230V, 50Hz AC supply. Calculate: (1) Inductive & capacitive reactance, (2) Total impedance, (3) Current and power factor, (4) Active and reactive power.

> 📊 **Question Schematic / State Specification:**

```
Question Series RLC Schematic: 230V, 50Hz ➔ R (10Ω) ➔ L (0.1H) ➔ C (50μF).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

Step-by-Step Impedance & Power Calculations:

- **1. Reactances:**

$$X_L = 2\pi f L = 2 \times 3.1416 \times 50 \times 0.1 = \mathbf{31.42\ \Omega}$$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2 \times 3.1416 \times 50 \times (50 \times 10^{-6})} = \mathbf{63.66\ \Omega}$$

$$\text{Net Reactance: } X = X_L - X_C = 31.42 - 63.66 = \mathbf{-32.24\ \Omega} \quad \textbf{(Capacitive!)}$$

- **2. Circuit Impedance:**

$$|Z| = \sqrt{R^2 + X^2} = \sqrt{10^2 + (-32.24)^2} = \sqrt{100 + 1039.4} = \mathbf{33.76\ \Omega}$$

- **3. Circuit Current:**

$$I = \frac{V}{|Z|} = \frac{230}{33.76} = \mathbf{6.813}\ \text{A}$$

- **4. Power Factor & Power Calculations:**

$$\text{Power Factor: } \cos\phi = \frac{R}{|Z|} = \frac{10}{33.76} = \mathbf{0.2962} \quad \textbf{(Leading)}$$

$$\text{Active Power: } P = V I \cos\phi = 230 \times 6.813 \times 0.2962 = \mathbf{464.15}\ \text{W}$$

$$\text{Reactive Power: } Q = V I \sin\phi = 230 \times 6.813 \times \sqrt{1 - 0.2962^2} = \mathbf{1494.5}\ \text{VAR (Leading)}$$

Because capacitive reactance dominates (`XC > XL`), the circuit draws a leading current of 6.81 A at a power factor of 0.296 leading.

🎯 **Verified Result:** Question 17 step-by-step resolution per MMDU university marking scheme.

📘 Question 21 (6 Marks): Superposition and Thevenin Verification on Multi-Source Network

Question:

For a network with two DC voltage sources V_1 = 30V (with internal resistance 2Ω) and V_2 = 20V (with internal resistance 4Ω) connected in parallel across a common load resistor RL = 8Ω. Find current through RL using: (1) Superposition Theorem, (2) Thevenin's Theorem. Verify both methods give identical results.

> 📊 **Question Schematic / State Specification:**

```
Question Parallel Source Bridge Topology: Branch 1 (30V, 2Ω) &parallel; Branch 2 (20V, 4Ω) ➔ Load RL (8Ω).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

Solution Step-by-Step Dual Theorem Verification:

`Method 1: Superposition Theorem`

`Case A: $V_1$ (30V) active, $V_2$ shorted:`

`$R_eq,_1 = 2 + (4 \parallel 8) = 2 + \frac{32}{12} = 2 + 2.667 = 4.667\ \Omega \Rightarrow I_1 = 30 / 4.667 = 6.429\ A$`

`Current through RL: $IL_1 = 6.429 \times [\ 4 / (4 + 8)\ ] = 6.429 \times \frac{4}{12} = 2.143\ A$ (↓)`

`Case B: $V_2$ (20V) active, $V_1$ shorted:`

`$R_eq,_2 = 4 + (2 \parallel 8) = 4 + \frac{16}{10} = 5.60\ \Omega \Rightarrow I_2 = 20 / 5.60 = 3.571\ A$`

`Current through RL: $IL_2 = 3.571 \times [\ 2 / (2 + 8)\ ] = 3.571 \times \frac{2}{10} = 0.714\ A$ (↓)`

`Total Load Current: $IL = IL_1 + IL_2 = 2.143 + 0.714 = 2.857\ A$`

`Method 2: Thevenin's Theorem`

`Open circuit terminals of RL: $V_{th} = V_2 + (V_1 - V_2) \times [\ 4 / (2 + 4)\ ] = 20 + 10 \times \frac{4}{6} = 26.67\ V$`

`$R_{th} = 2 \parallel 4 = \frac{2 \times 4}{2 + 4} = 8 / 6 = 1.333\ \Omega$`

`Load Current: $IL = V_{th} / (R_{th} + RL) = 26.67 / (1.333 + 8) = 26.67 / 9.333 = 2.857\ A$`

Both Superposition and Thevenin theorems converge to an identical load current of **2.857 A**.

🎯 **Verified Result:** Question 21 step-by-step resolution per MMDU university marking scheme.